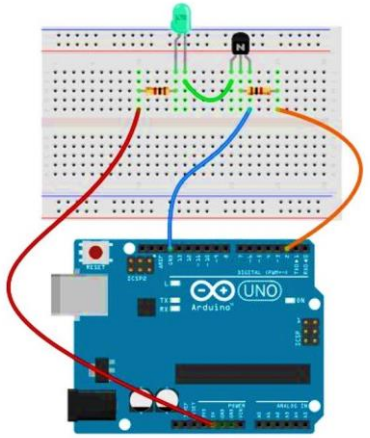
Introduction to *the* Bipolar Junction Transistor
The Common Emitter Circuit

EE 312 EXPERIMENT 9

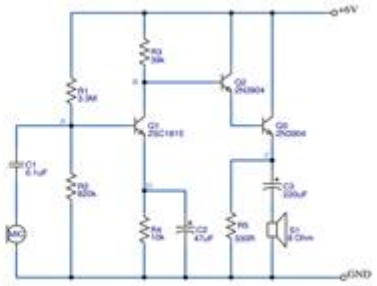
Where Do We Need a Transistor?

- To *electronically* Switch ON/OFF a Load That Needs Values of Current and/or Voltage Not Available from a Controlling Signal Source.

- To Amplify a Small Signal (Example: Audio Amplifier)





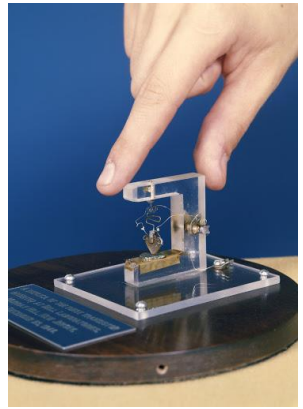


BJT Name Origin and History

Bipolar (two types of carriers) Junction (two p-n junctions) Transistor

Invented at Bell Labs by William Shockley, Walter Brattain, and John Bardeen so late in 1947.

The BJT is more popular in discrete-circuit design, automotive applications, radio frequency (RF) circuits, and very high-speed digital circuits.



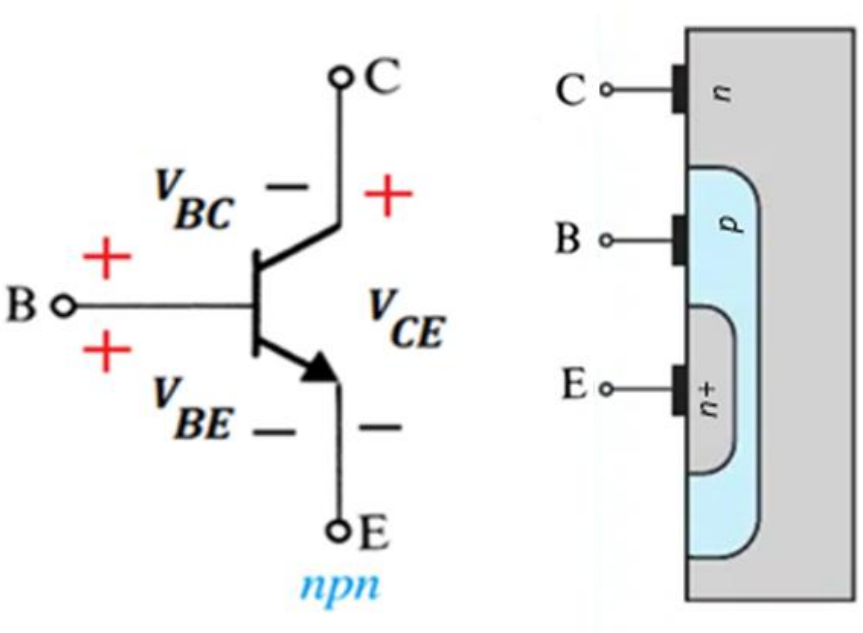


Structure of the npn BJT

Middle layer *p*-type [Base].

Heavily doped *n*-type layer [Emitter].

Lightly doped *n*-type layer [Collector].



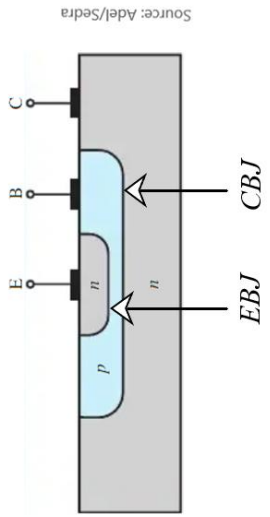
The BJT is NOT Symmetrical

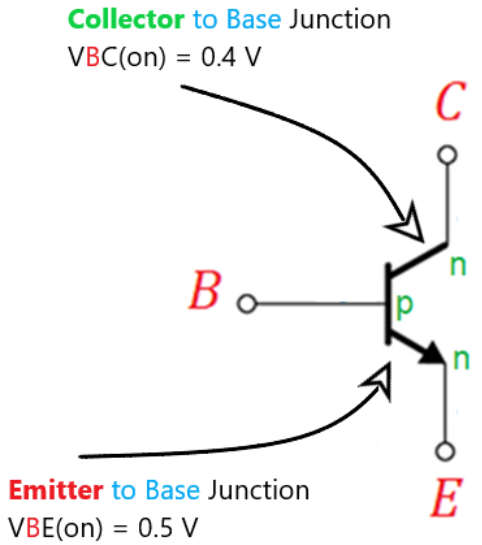
The CBJ has a greater cross sectional area than the EBJ

⇒ the CBJ has a smaller turn-on voltage

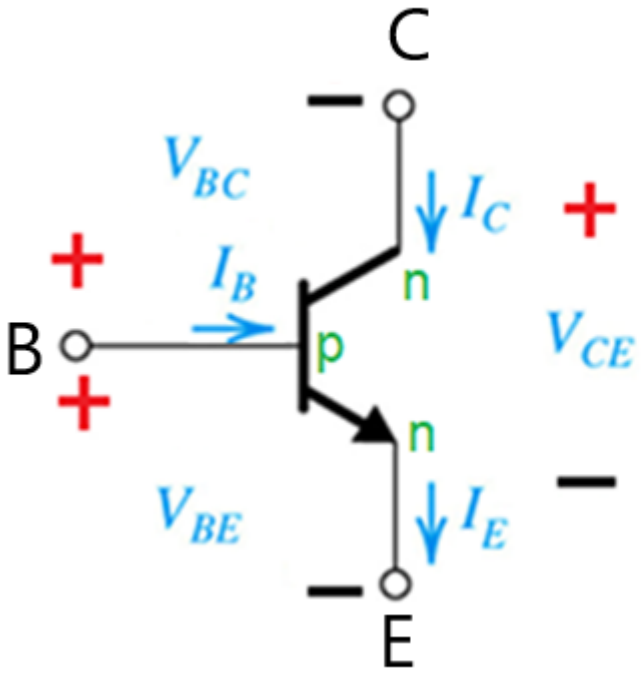
$V_{BC(on)} \approx 0.4$ V

$V_{BE(on)} \approx 0.5$ V





KVL and KCL DO Apply to a BJT

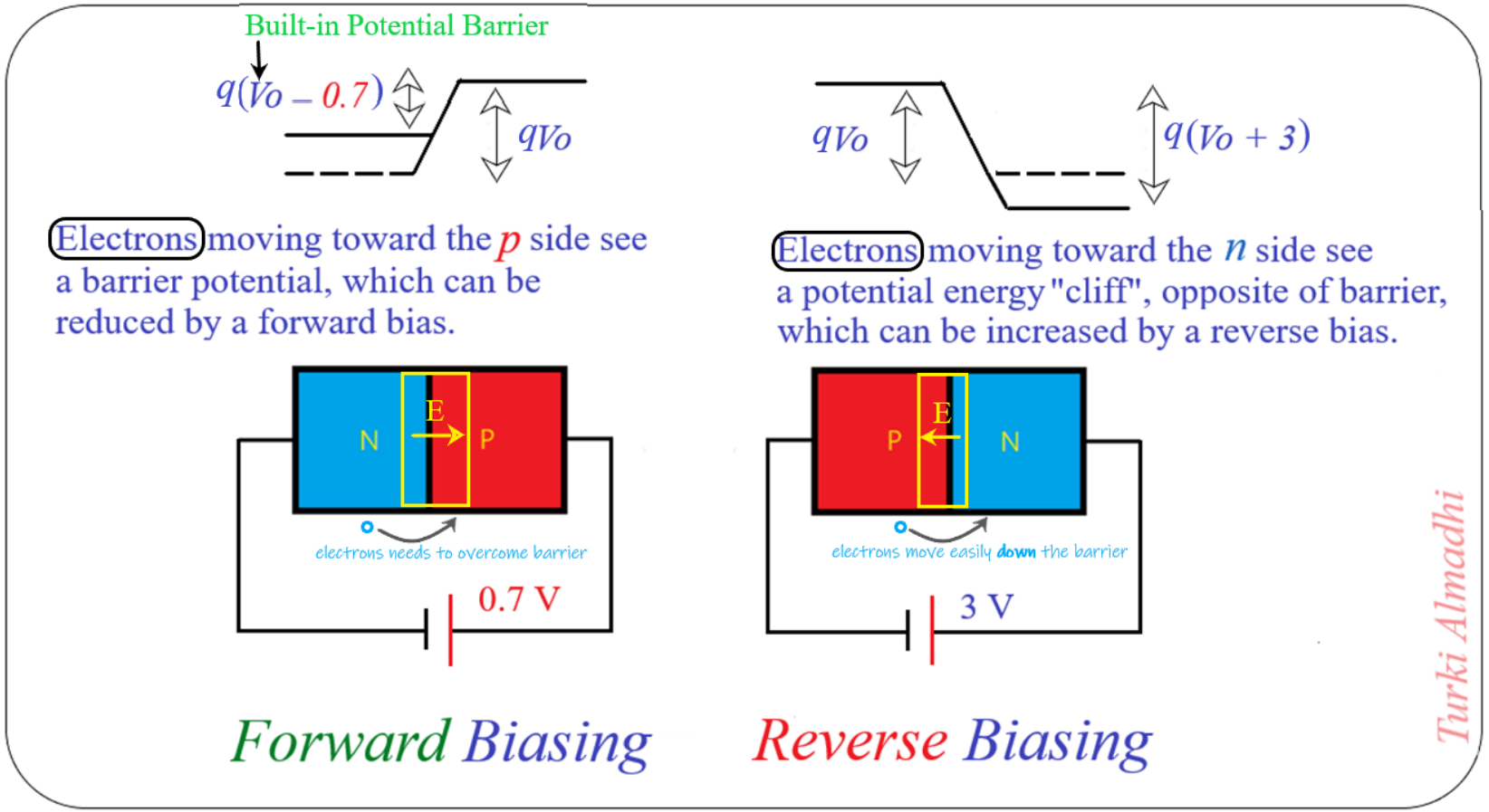


$$I_E = I_C + I_B$$ by KCL

$$V_{CE} = V_{CB} + V_{BE}$$ by KVL

$$V_{CE} = V_{BE} - V_{BC}$$

Revision of the PN Junction Bias Conditions



The Transistor Action!

In the Active Mode of Operation

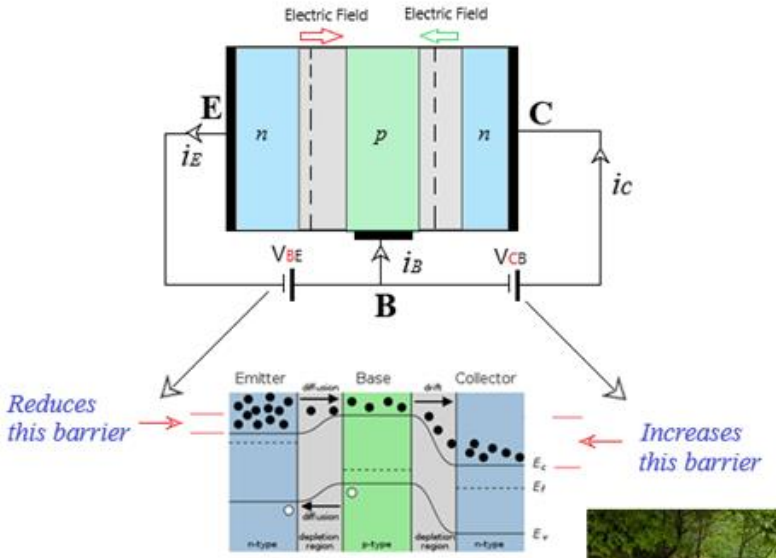
VBE > 0.5 V forward biases the EBJ ⇒ lowers the potential barrier ⇒ electrons are injected from the emitter into the base.

The base width is small enough ⇒ most electrons diffuse to reach the CBJ, where the are collected by the collector, provided VBC < 0.4 to insure efficient collection.

Since V_{BE} controls I_E and I_C is proportional to I_E ($I_C = \alpha I_E$) ⇒

a transistor action is obtained!

Electrons *need* energy to go *in the direction of* an electric field, but *loose* energey if they move *against* one.

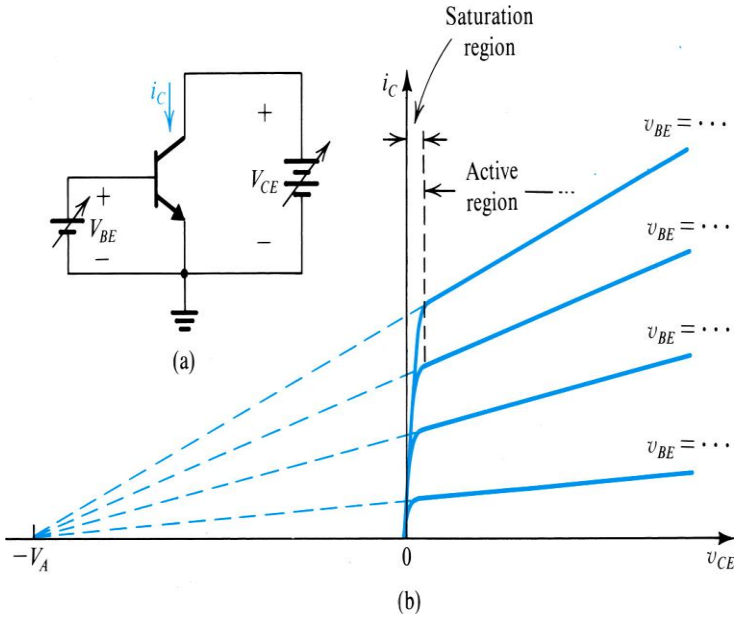




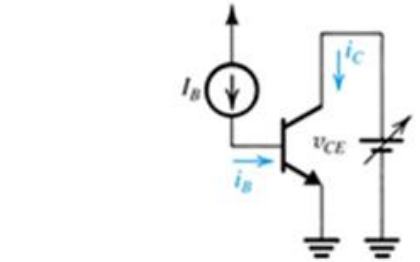
i_C vs. υ_{CE} Terminal Characteristics

VCCS (left), CCCS (right)

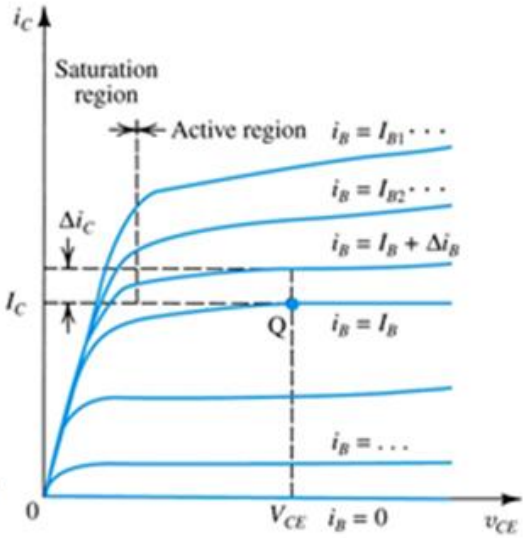
IN THE ***ACTIVE*** REGION, I_C IS CONTROLLED EXPONENTIALLY BY V_{BE}.



IN THE ***ACTIVE*** REGION, THE RELATIONSHIP BETWEEN I_C AND I_B IS LINEAR.



Changing I_B changes V_{BE}



Summary of Modes of Operation for the npn BJT

MODE	EBJ Bias Condition	CBJ Bias Condition	Circuit Model	Application
Cutoff	VBE < VBE(on) Cutoff	VBC < VBC(on) Cutoff	*Open switch* between the collector and the emitter	Switching Circuits, like logic gates
Saturation	VBE ≥ VBE(on) Forward Biased	VBC ≥ VBC(on) Forward Biased	*Closed switch* (with offset of about 0.2 V) between the collector and the emitter	Switching Circuits, like logic gates
Active	VBE ≥ VBE(on) Forward Biased	VBC < VBC(on) Cutoff	***A PN junction*** between the base and the emitter and a ***Voltage-Controlled Current Source*** between the collector and the emitter	Amplifier
VBE(on) ≈ 0.5 V and VBC(on) ≈ 0.4 V				

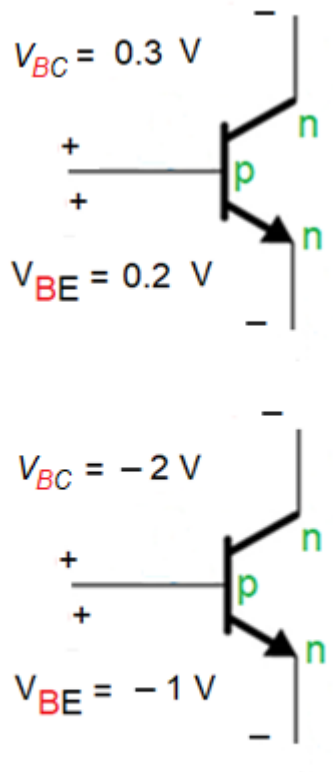
The Cutoff/Open-Switch Mode

A npn BJT is said to be in the cutoff mode if:

$$VBE < VBE(on) \approx 0.5\ V$$

and

$$VBC < VBC(on) \approx 0.4\ V$$



Examples of BJTs in Cutoff Mode

The Saturation/Closed-Switch Mode

An npn BJT is said to be in the saturation mode if

$V_{BE} \geq V_{BE(on)} \approx 0.5$

Typically $V_{BE} \approx 0.7$ V

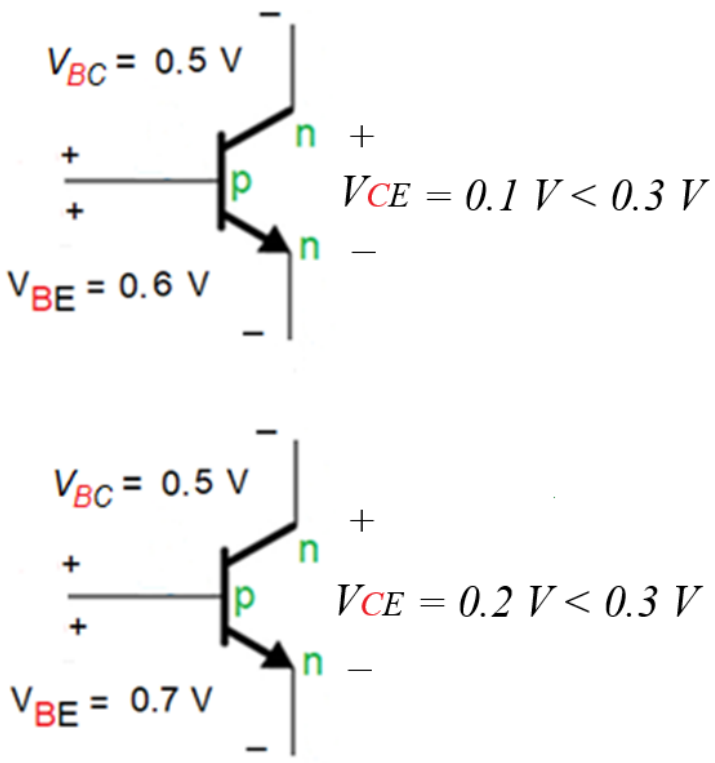
and

$V_{BC} \geq V_{BC(on)} \approx 0.4$ V

$\Downarrow$

$$V_{CEsat} = 0.1\text{–}0.2\text{V}$$

Also, $I_{Csat} / I_B < \beta_{min}$



Examples of BJTs in Saturation

The Active Mode (Employed in Amplifiers)

An npn BJT is said to be in the active mode if

$V_{BE} \geq V_{BE(on)} \approx 0.5$

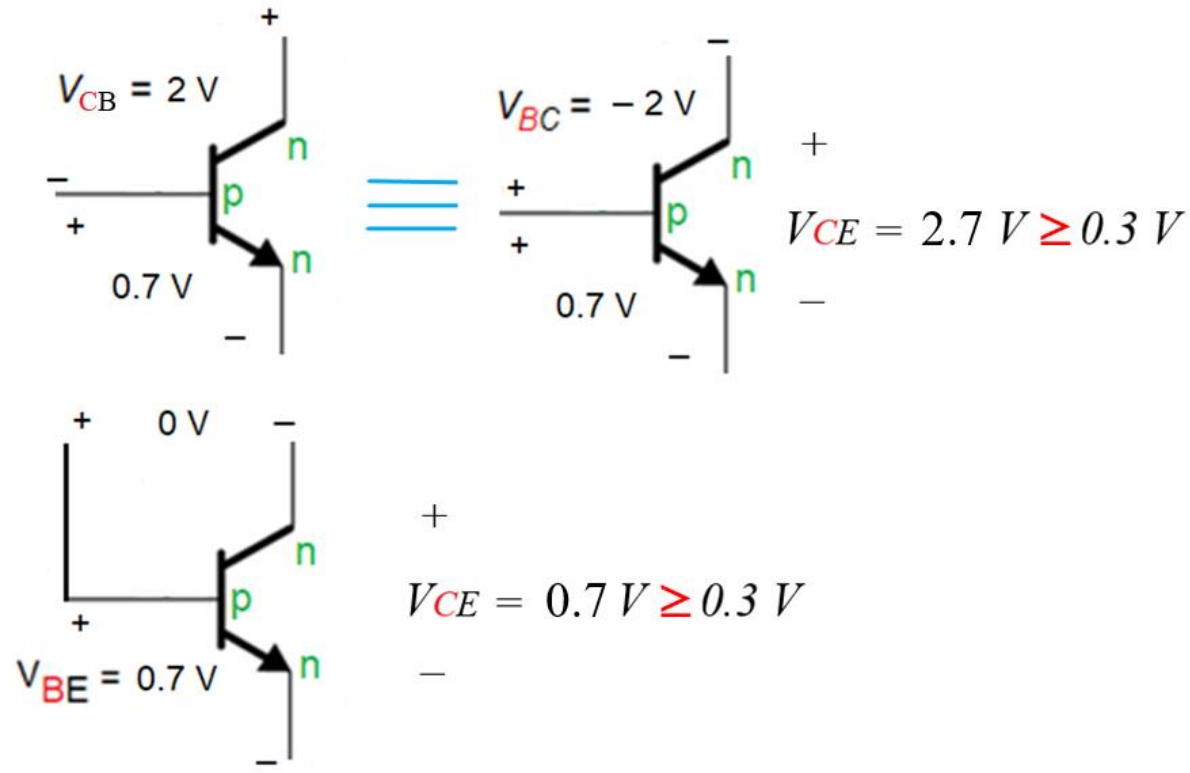
Typically $V_{BE} \approx 0.7$ V

and

$V_{BC} < V_{BC(on)} \approx 0.4$ V

$\Downarrow$

$V_{CE} \geq 0.3$ V



Examples of BJT in the Active Mode

Ratio of I_C to I_B in the *Saturation* and *Active* Modes

By definition, in the **saturation** mode:

$$I_{Csat} / I_B \equiv \beta_{forced}$$

β_{forced} can be controlled by the circuit designer.

By definition, in the **active** mode:

$$I_C / I_B \equiv \beta$$

β is a BJT parameter, it is denoted h_{FE} in datasheets

$$\beta_{forced} < \beta_{min}$$

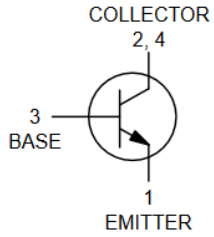
BD137 G Datasheet (1/2)

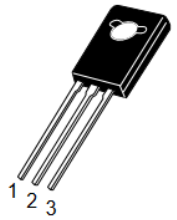
- High DC Current Gain
- BD 135, 137, 139 are complementary with BD 136, 138, 140
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant*

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector−Emitter Voltage BD135G BD137G BD139G	V_{CEO}	 45 60 80	Vdc
Collector−Base Voltage BD135G BD137G BD139G	V_{CBO}	 45 60 100	Vdc
Emitter−Base Voltage	V_{EBO}	5.0	Vdc
Collector Current	I_C	1.5	Adc
Base Current	I_B	0.5	Adc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	 1.25 10	Watts mW/°C
Total Device Dissipation @ T_C = 25°C Derate above 25°C	P_D	 12.5 100	Watts mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	−55 to +150	°C

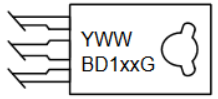
1.5 A POWER TRANSISTORS
NPN SILICON
45, 60, 80 V, 12.5 W





TO−225
CASE 77−09
STYLE 1

MARKING DIAGRAM



BD137 G Datasheet 2/2

BD135G, BD137G, BD139G

ELECTRICAL CHARACTERISTICS (T_C = 25°C unless otherwise noted)

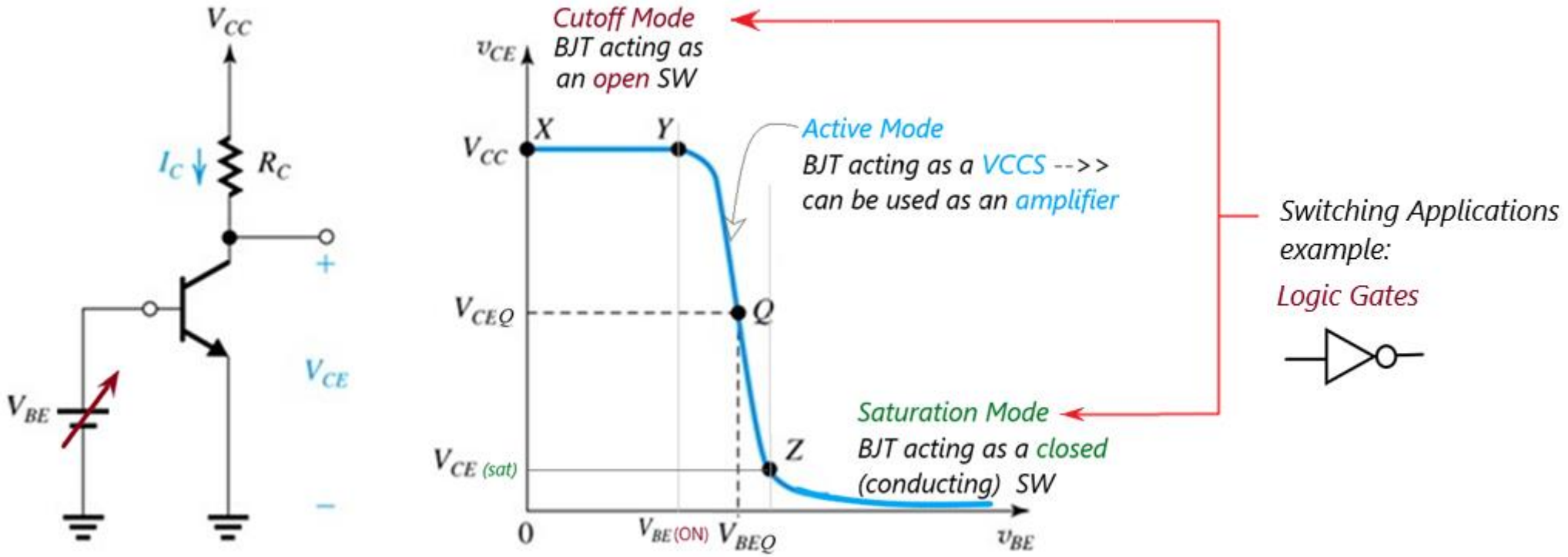
Characteristic	Symbol	Min	Max	Unit
Collector−Emitter Sustaining Voltage* (I_C = 0.03 Adc, I_B = 0) BD135G BD137G BD139G	BV_{CEO}*	 45 60 80	 – – –	Vdc
Collector Cutoff Current (V_{CB} = 30 Vdc, I_E = 0) (V_{CB} = 30 Vdc, I_E = 0, T_C = 125°C)	I_{CBO}	 – –	 0.1 10	μAdc
Emitter Cutoff Current (V_{BE} = 5.0 Vdc, I_C = 0)	I_{EBO}	–	10	μAdc
DC Current Gain (I_C = 0.005 A, V_{CE} = 2 V) (I_C = 0.15 A, V_{CE} = 2 V) (I_C = 0.5 A V_{CE} = 2 V)	h_{FE}*	 25 40 25	 – 250 –	–
Collector−Emitter Saturation Voltage* (I_C = 0.5 Adc, I_B = 0.05 Adc)	$V_{CE(sat)}$*	 –	 0.5	Vdc
Base−Emitter On Voltage* (I_C = 0.5 Adc, V_{CE} = 2.0 Vdc)	$V_{BE(on)}$*	 –	 1	Vdc

Beta = IC/IB

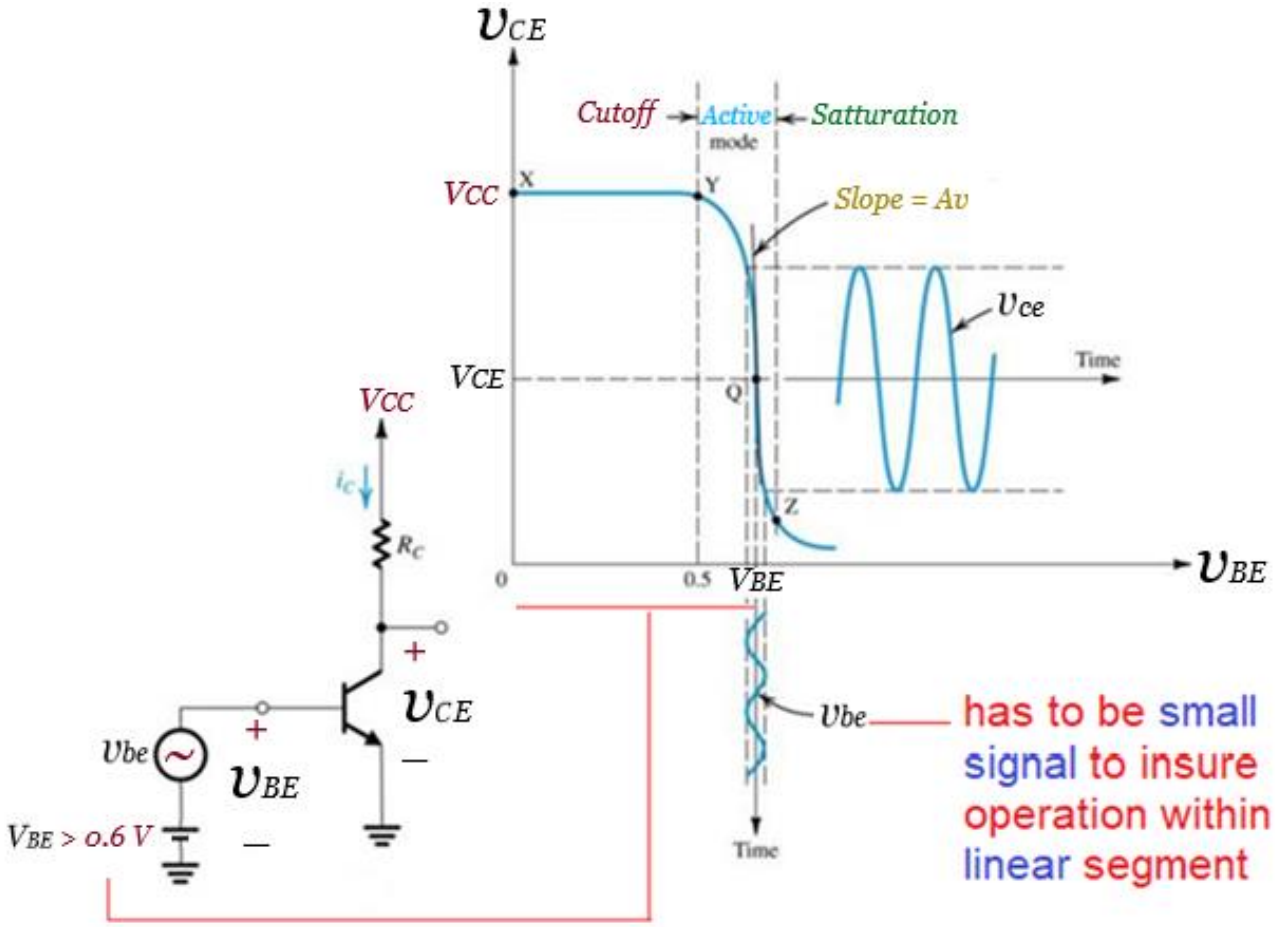
Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

*Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

The Common Emitter circuit Voltage Transfer Characteristic



Using The CE Circuit As an Amplifier



Switching Application Design (1/2)

~ 0.2 V

$$R_C = \frac{V_{CC} - V_{LED} - V_{CEsat}}{I_{Csat}}$$

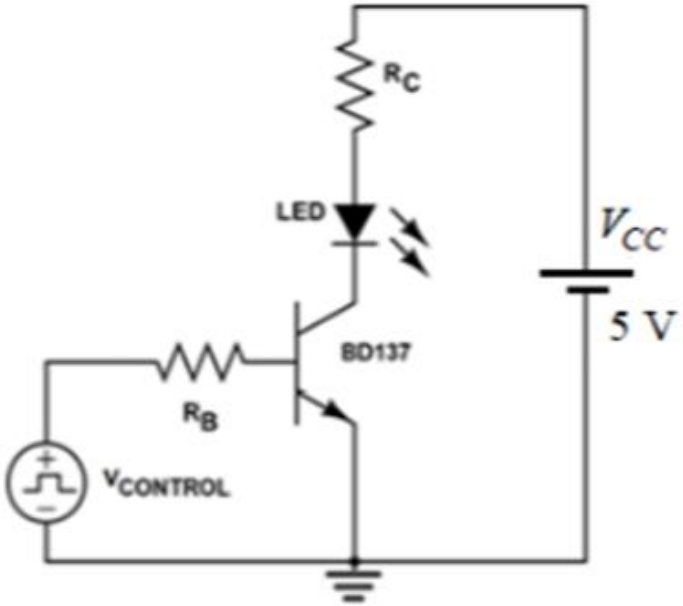
The same as the load's current; in *this* case the *targeted* *LED* current

$$I_B = \frac{I_{Csat}}{\beta_{forced}}$$

(Note: *Select* $\beta_{forced} < \beta_{min}$)

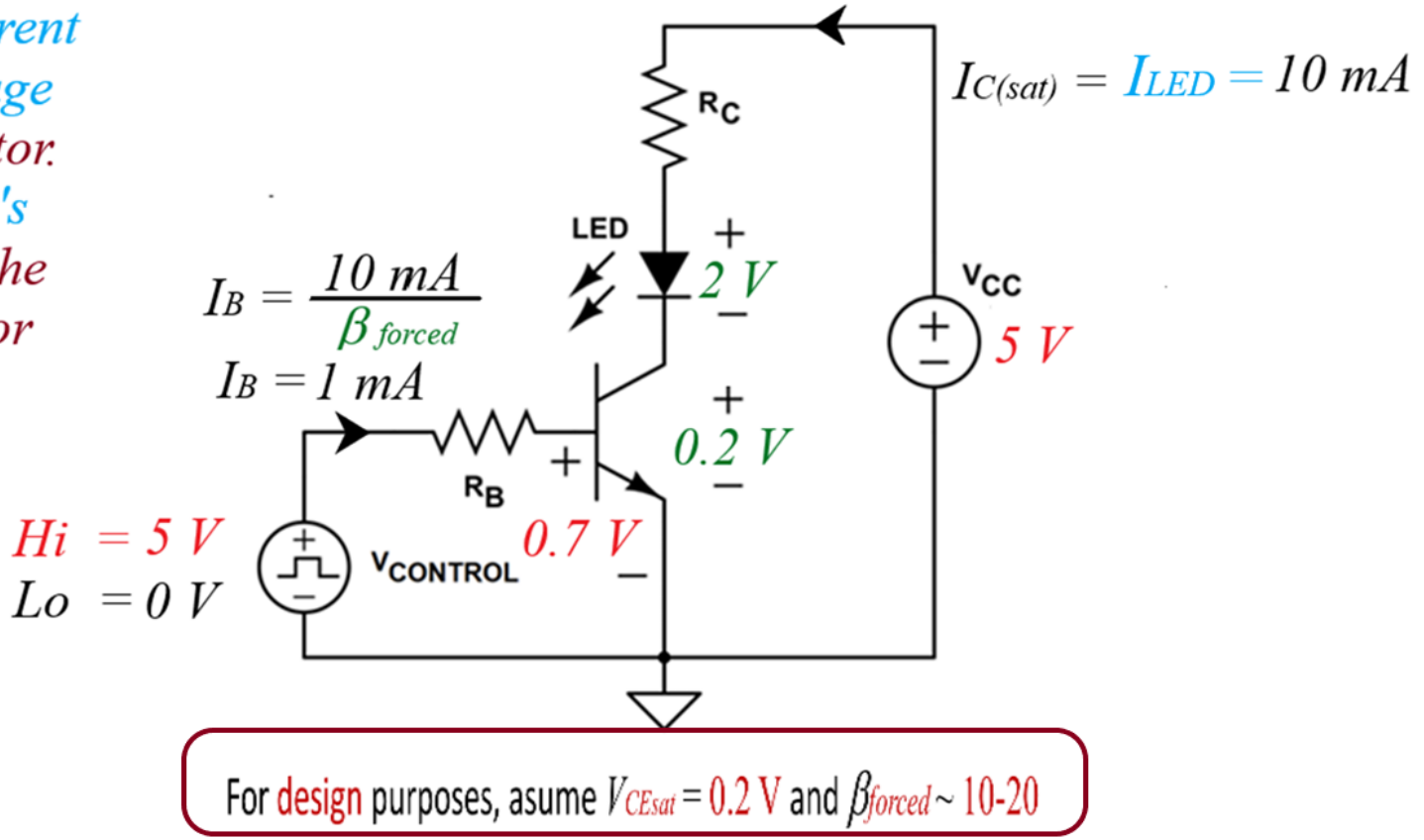
$\beta_{forced} \sim 10\text{-}20$ ← Rule of thumb for design purposes ONLY!

$$R_B = \frac{V_{Contol_Hi} - 0.7}{I_B}$$



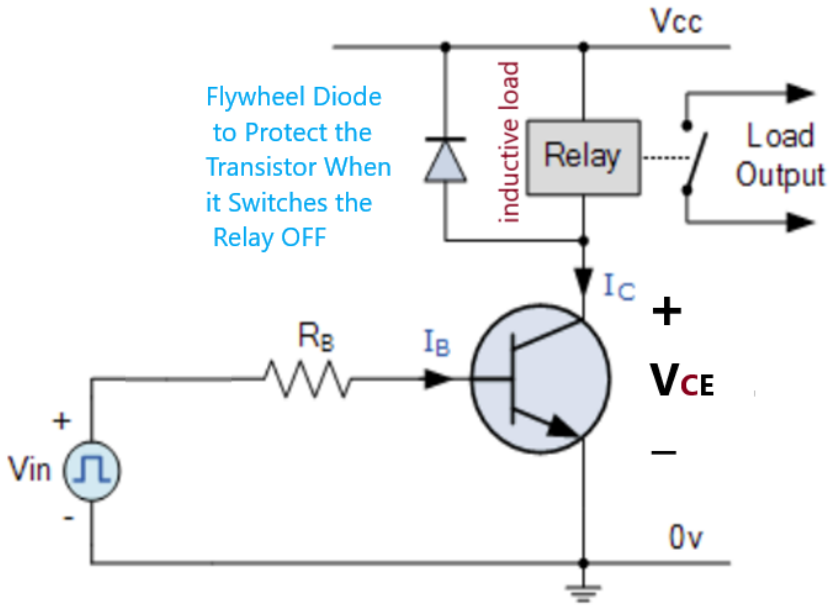
Switching Application Design (2/2)

You know the current into and the voltage across each resistor. You can use Ohm's low to calculate the required values for each resistor.



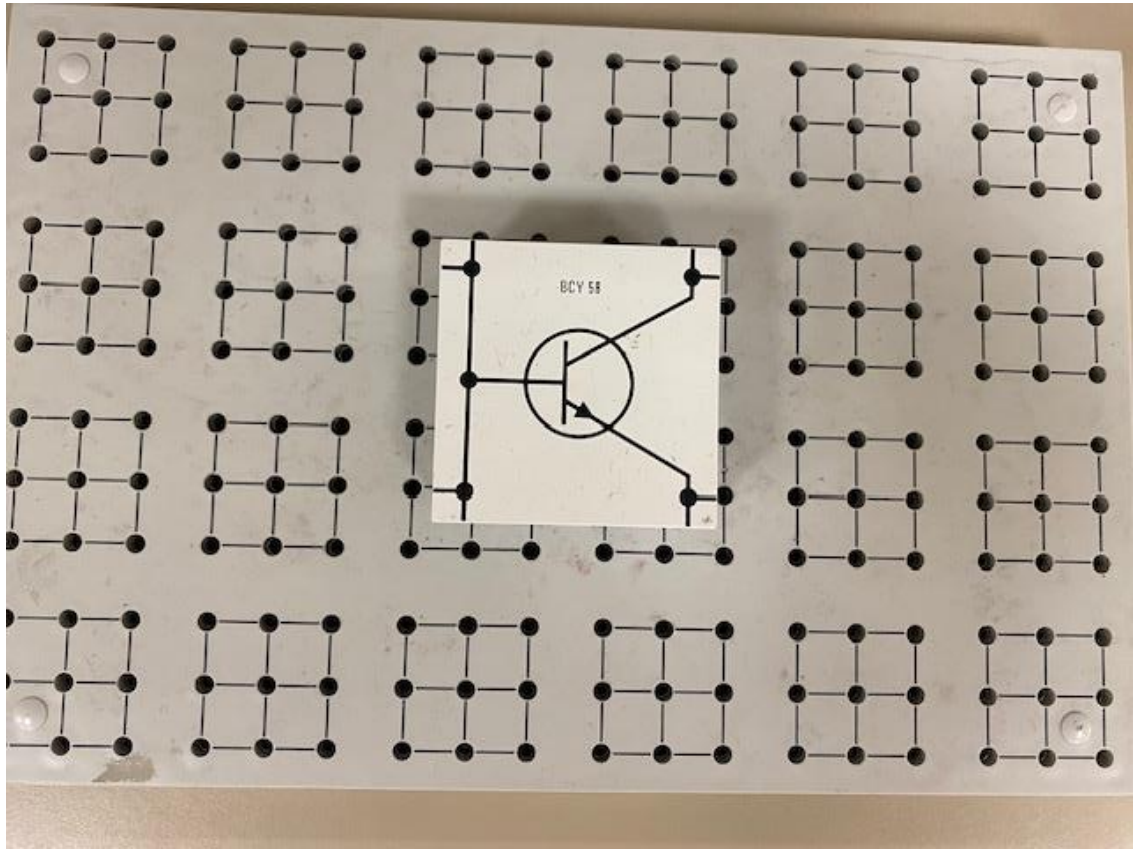
Switching an Inductive Load

The Need for a Flywheel Diode



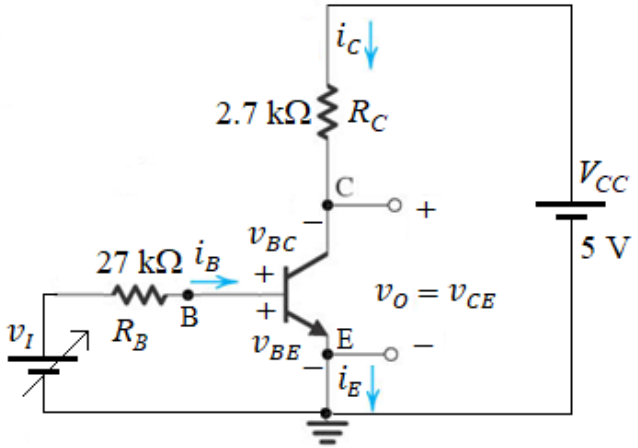
Wnen the Transistor Switches the Relay off ,
Large Inductive VCE Might Damage the Transistor!
To Avoid that, Use a Flywheel Diode to Limit VCE to VCC + 0.7 V.

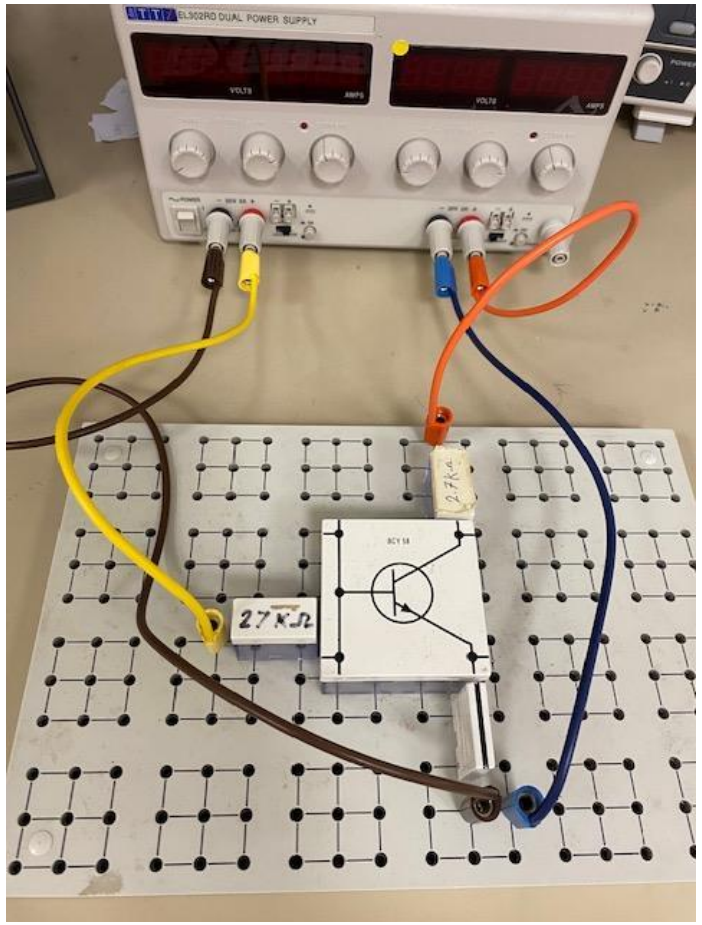
The BJT On the Socket Board





Setup For Procedure A

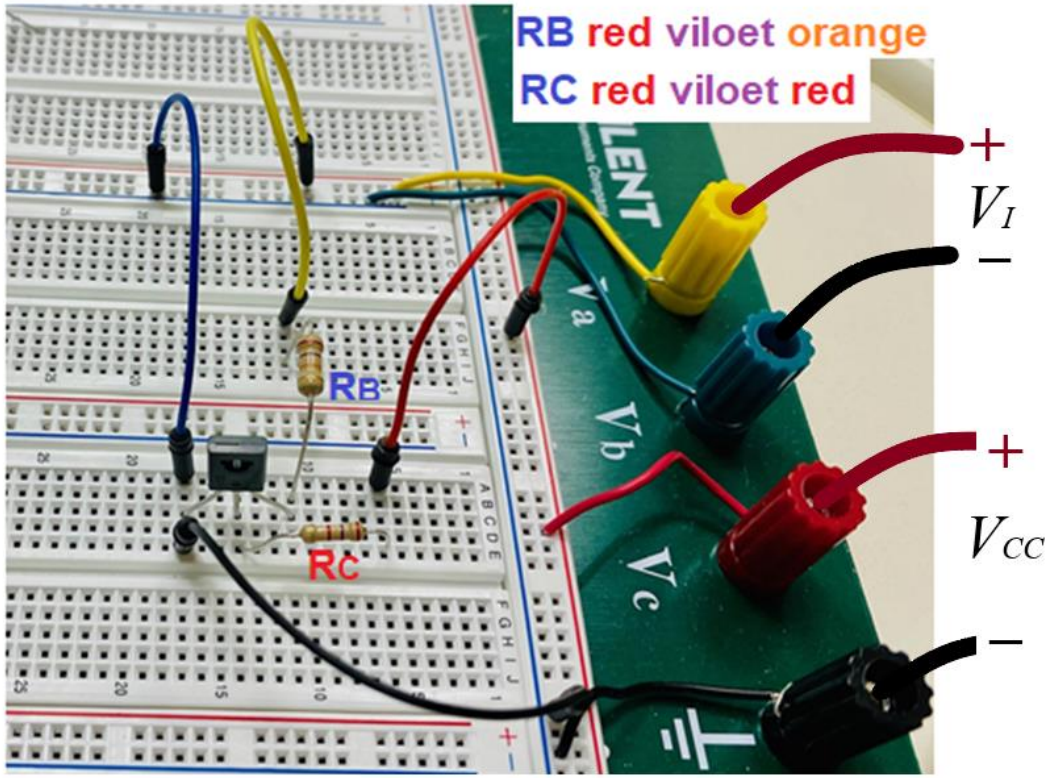




Breadboard Revision



Circuit Diagram and Breadboard Implementation



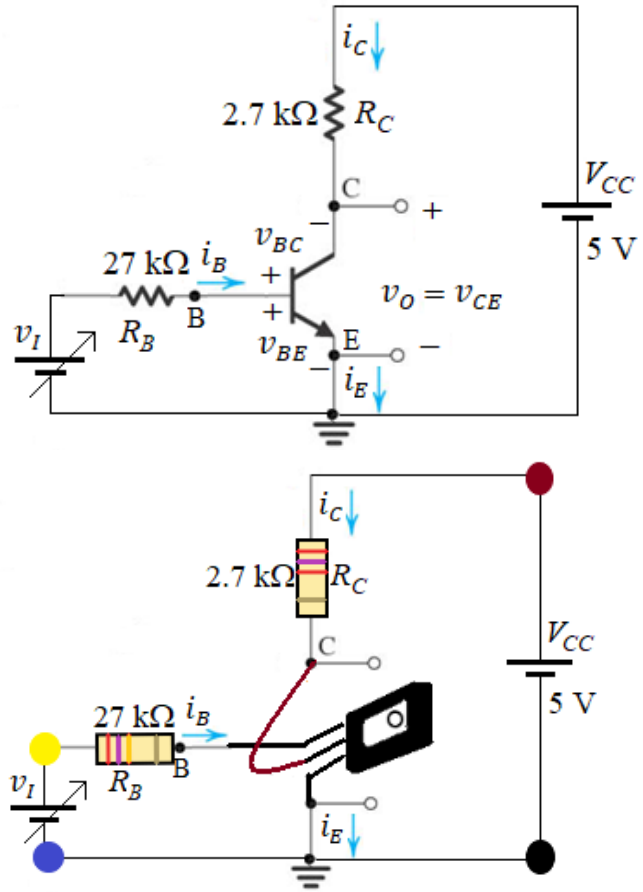


Table to Store Measured Data

Vary V_I to obtain ↓	To be Measured		To be calculated				
$V_{CE} = V_O$ (V)	V_{BE} (V)	V_I (V)	$V_{BC} = V_{BE} - V_{CE}$ (V)	Mode of operation	I_B (mA)	I_C (mA)	I_E (mA)
5.000	0.000	0.000	-5.000	Cutoff	0.000	0.000	0.000
4.950							
4.500							
3.500							
2.500							
2.000							
1.500							
0.500							
0.200							
0.100							

References

A. S. Sedra and K. C.Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.

T. A. Almadhi, EE 312 Lab Manual: https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/EE312.pdf